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Spatial8 Studio: A Web-Based Platform for Dynamic Binaural Audio Spatialization Without HRTF Databases

Avulla Utej
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Correspondence: spatial8studio@gmail.com | DOI: 10.1186/spatial8-binaural-2023

ABSTRACT

We present Spatial8 Studio, a web-based platform for converting ordinary mono and stereo audio into perceptually convincing binaural earphone audio without requiring measured Head-Related Transfer Function (HRTF) databases. While prior platforms such as PlugSonic (Comunita et al., 2022) demonstrated the value of web-accessible binaural audio tools, they relied on third-party spatialization toolkits and generic HRTF convolution. Spatial8 instead implements a lightweight, physics-grounded 2-factor structural model Interaural Time Difference (ITD) via Woodworth spherical head geometry and Interaural Level Difference (ILD) via a first-order continuous head-shadow biquad filter to spatially anchor audio sources along the horizontal plane. The platform further introduces multi-track scene mixing with automatic sample rate harmonization, constant-power panning, and phase-coherent stereo downmixing. Informal evaluations with non-specialist participants confirm clear, stable left-right localization with no HRTF-induced spectral coloration artifacts. Keywords: binaural audio, spatial sound, ITD, ILD, web audio platform, earphone spatialization, multi-track mixing, structural binaural synthesis.

I. INTRODUCTION

Binaural audio the technique of reproducing three-dimensional spatial hearing through standard stereo headphones has historically been confined to specialist audio engineers with access to anechoic chambers, acoustic manikins, and proprietary Digital Audio Workstation (DAW) plugins. The emergence of web audio technologies has begun to democratize spatial audio production, as demonstrated by platforms such as PlugSonic [1], which enable non-specialist users to curate, position, and experience 3D soundscapes through standard browsers and mobile devices without specialized hardware.

Despite these advances, most web-based binaural audio platforms inherit a critical dependency: they rely on generic Head-Related Transfer Function (HRTF) databases measured from acoustic manikins or human subjects. When applied to listeners whose ear geometry differs from the measurement subject, generic HRTF convolution causes well-documented spectral coloration artifacts, timbre distortion, and front-back confusion [2,3]. This dependency also imposes significant computational overhead each spatialization requires convolving the audio signal with long FIR filters (typically 256-512 taps per ear) for every virtual source position.

Spatial8 Studio addresses these limitations through a different architectural philosophy: rather than convolving with empirical HRTF tables, it synthesizes the dominant horizontal binaural cues analytically using a physics-grounded structural model [4]. The result is a web-accessible platform that delivers perceptually effective left-right spatialization with zero HRTF dependency, O(1) per-sample computation, and no coloration artifacts in the horizontal plane the acoustic dimension most relevant to music, gaming, voice communication, and virtual soundscape design.

This paper makes the following contributions: (1) a physics-based binaural synthesis engine for the web without HRTF databases; (2) multi-track scene mixing with automatic sample rate harmonization; (3) phase-coherent stereo-to-mono downmixing for heterogeneous audio sources; and (4) a usability evaluation with non-specialist participants confirming effective perceptual lateralization.

II. RELATED WORK

A. Web-Based Binaural Audio Platforms

The most closely related work is PlugSonic [1], a web and mobile platform for dynamic and navigable binaural audio developed under the EU PLUGGY cultural heritage project. PlugSonic comprises two complementary tools: PlugSonic Sample, a browser-based audio editor for effects processing, and PlugSonic Soundscape, an interactive environment for positioning virtual sound sources on a 2D map and experiencing them on mobile devices via ARKit-based head tracking. For binaural rendering, PlugSonic delegates to the 3D Tune-In Toolkit (3DTI), a third-party open-source library that applies measured HRTF filters. PlugSonic demonstrated that everyday users and cultural institutions can engage with spatial audio design without specialized audio engineering backgrounds a democratization goal we share with Spatial8 Studio.

However, PlugSonic's reliance on the 3DTI toolkit introduces HRTF-related coloration artifacts for listeners whose ear geometry differs from the measurement manikin. Its architecture also requires WebAssembly bindings to the native 3DTI C++ library, adding deployment complexity. Spatial8 avoids both issues by synthesizing binaural cues analytically, with no external toolkit dependency.

B. Structural Binaural Synthesis Models

The foundational acoustic model underlying Spatial8 is the structural binaural synthesis framework of Brown and Duda (1998) [4], which decomposes the binaural transfer function into modular, physics-based components: (1) Woodworth's spherical head Interaural Time Difference (ITD) formula and (2) a continuous first-order pole-zero head-shadow filter for Interaural Level Difference (ILD). Unlike HRTF convolution, this model requires no measurement data and provides continuous, artifact-free parameter variation as the source azimuth changes. Brown and Duda experimentally verified that this 2-factor model accounts for over 95% of human horizontal localization acuity.

Earlier models by Woodworth (1938) [5] and Kuhn (1977) [6] established the geometric basis for ITD estimation from spherical head approximations. Shaw (1974) [7] characterized the outer ear transfer function. Rayleigh's Duplex Theory (1907) [8] identified ITD as the dominant cue below 1.5 kHz and ILD above, providing the theoretical backbone for the 2-factor horizontal model.

C. Limitations of Existing Approaches

- HRTF Personalization Problem: Generic HRTFs measured from KEMAR or similar manikins embed spectral notches specific to that ear geometry. Applied to a different listener, these produce timbre coloration perceived as muffling rather than spatial externalization [2].
- Computational Overhead: Real-time convolution with 256-512 tap FIR filters per source per ear at 44.1 kHz demands significant CPU resources, limiting concurrent source count on consumer devices.
- Static Source Grids: Most HRTF databases (e.g., CIPIC [9]) tabulate filters at discrete angular grid positions (typically 5-degree increments). Moving a source between grid positions requires FIR coefficient interpolation, causing audible artifacts including phase smearing and zipper noise.
- Elevation vs. Horizontal Trade-off: Full 3D spatialization requiring elevation cues mandates listener-specific pinna HRTFs. Without personalization, elevation rendering degrades dramatically. Spatial8 deliberately restricts spatialization to the horizontal plane where the structural model is physically sufficient.

III. BINAURAL AUDIO SIGNAL PROCESSING

A. The Structural Binaural Model

Spatial8 Studio implements the two-factor structural model of Brown and Duda [4]. The human head is approximated as a rigid sphere of radius a = 0.0875 m (the average adult head radius). For a sound source at azimuth angle theta (radians, measured from the frontal midline), the Interaural Time Difference delivered to the contralateral (far) ear is given by Woodworth's formula:

```
tau(theta) = (a/c) * (theta + sin theta),   |theta| <= pi/2
```

a = 0.0875 m, c = 343 m/s; at theta = pi/2 (full lateral): tau = 0.655 ms ~= 29 samples at 44.1 kHz

The contralateral ear signal is delayed by tau(theta) samples. The ipsilateral (near) ear receives the signal with zero additional delay. This single delay introduces the dominant low-frequency lateralization cue exploited by the human auditory system below 1.5 kHz (Rayleigh's Duplex Theory).

B. The Head-Shadow ILD Filter

At frequencies above 1.5 kHz, the rigid skull casts an acoustic shadow over the contralateral ear. Brown and Duda [4] derived a minimum-phase first-order continuous-time transfer function that approximates this head-shadow diffraction effect:

```
H_hs(s, theta) = [1 + alpha(theta)*(s/omega_0)] / [1 + (s/omega_0)]
```

omega_0 = c/a = 3920 rad/s; alpha(theta) = 1 + cos(theta), smoothly 2.0 -> 0.0 as theta: 0 -> pi

Applying the bilinear transform converts this to a first-order IIR digital filter requiring only two multiplications and one state register per audio sample. This filter introduces a natural high-shelf attenuation of 8-15 dB at frequencies above 2 kHz on the shadowed ear. Critically, because alpha(theta) varies continuously with azimuth, source motion produces smooth, artifact-free ILD changes with no filter switching clicks.

C. Constant-Power Amplitude Panning

In addition to the physics-based ITD and ILD, Spatial8 applies constant-power (sine/cosine) amplitude panning to maintain perceptually uniform loudness across the soundstage. For a normalized pan position p in [-1, +1]:

```
gainL = cos((p+1)*pi/4),   gainR = sin((p+1)*pi/4)
```

ensures gainL^2 + gainR^2 = 1.0 (constant total power) at all pan positions

D. Stereo Input Handling and Phase Coherence

Many real-world audio files are stereo (two-channel). Naive L+R averaging can introduce comb filtering in out-of-phase stereo content. Spatial8 computes the normalized cross-channel sample correlation before downmixing: if the correlation exceeds 0.7, channels are summed and averaged; otherwise the mid signal M = (L+R)/2 is extracted, discarding the side signal to prevent phase anomalies. This ensures the input to the spatialization chain is a clean, phase-coherent mono signal regardless of the original recording format.

E. Multi-Source Scene Mixing and Sample Rate Harmonization

Real-world multi-track sessions mix stems recorded at different sample rates (44.1 kHz, 48 kHz, 96 kHz). Spatial8 resamples all stems to a common target rate using linear interpolation before spatialization. When K audio sources are placed at azimuth angles theta_1 ... theta_K, the binaural

output follows the acoustic Principle of Superposition:

```
y_L[n] = SUM_{k=1}^{K} g_k * x_k[n] * h_L(theta_k),   y_R[n] = SUM_{k=1}^{K} g_k * x_k[n
```

where g_k is the per-track gain. The summed output is normalized by scaling to 98% of the peak sample value, maintaining maximum headroom while preventing inter-sample clipping peaks.

IV. PLATFORM ARCHITECTURE

A. Overview

Spatial8 Studio is a full-stack web platform structured across three layers: an interactive browser-based authoring environment for audio upload and spatial source placement; a server-side processing layer that orchestrates binaural rendering jobs; and a DSP processing engine that executes the structural binaural model. This architecture is conceptually analogous to PlugSonic's separation between its web curation interface and mobile experience layer [1], but consolidates both authoring and processing into a single web session, eliminating the need for a companion mobile application.

B. Audio Authoring Interface

The authoring interface presents a 2D spatial stage canvas mapping the horizontal soundstage from X = -1.0 (full left) to X = +1.0 (full right). Users drag virtual source markers to desired positions. Unlike PlugSonic's map-based coordinate system oriented toward physical location-aware experiences, Spatial8's stage canvas is designed for audio production workflows positioning instruments, voices, and sound effects within a musical or cinematic scene. Five arrangement presets accelerate common workflows: Stereo Pair (instruments at +/-0.6), Spread (four sources distributed evenly), Center (all sources at X=0), Cross Sweep, and Custom placement.

C. Server-Side Processing Layer

Audio files uploaded by the user are received by a REST API server, stored in a temporary staging area, and dispatched to the DSP engine. The server exposes two primary endpoints: one for single-file spatialization and one for multi-track scene rendering. After processing, the resulting binaural WAV file is streamed back to the browser for immediate playback and download. This client-server separation keeps the DSP computation off the browser's main thread, enabling arbitrarily complex DSP operations without impacting the authoring interface responsiveness.

D. DSP Processing Engine

The DSP engine is implemented in MATLAB and comprises two primary scripts. The single-source spatializer accepts a mono or stereo audio file and an azimuth parameter, applies the ITD delay and ILD head-shadow biquad, and writes a binaural WAV output. The multi-track renderer accepts N audio files with their respective coordinates, resamples all to a common sample rate, spatializes each independently, and linearly superposes the results with peak normalization. The MATLAB implementation benefits from vectorized array operations, making processing of full-length audio files (up to 10 minutes) computationally efficient without requiring real-time streaming architecture.

V. USER STUDY AND EVALUATION

A. Study Design

Following the evaluation methodology of Comunita et al. [1], we conducted a task-based informal listening evaluation with non-specialist participants. The study assessed three primary dimensions: (1) Spatialization Quality clarity and stability of left-right perceptual separation; (2) Coloration Assessment presence or absence of timbre changes attributable to the binaural processing; and (3) Usability ease of uploading audio, positioning sources on the spatial stage, and retrieving the processed output.

B. Participants and Procedure

Twelve participants (ages 19-34, 7 male, 5 female) with no professional audio engineering background were recruited. All used closed-back circumaural headphones (Sony WH-1000XM4 or equivalent) during evaluation. Each participant completed three tasks: (T1) upload a provided music sample and place it at X = -0.8 (far left), then assess left-right separation; (T2) create a 3-source scene with sources at X = -0.7, 0, and +0.7 and describe the perceived spatial arrangement; and (T3) rate whether the processed audio sounded colored or muffled compared to the unprocessed original.

C. Results: Spatialization Quality

All 12 participants (100%) correctly identified far-left placement in T1 without prompting. At intermediate positions (X = +-0.5), 10 of 12 (83%) perceived the source at a half-lateral position. In T2, 11 of 12 participants correctly described a left-center-right arrangement, with one participant reporting slight ambiguity between center and slight-right for the center source.

D. Results: Coloration Assessment

Zero participants reported timbre muffling, spectral coloration, or unnatural tonal change in T3. This contrasts with known HRTF coloration reports in the literature: Wenzel et al. (1993) [10] found that non-individual HRTFs caused front-back confusion in 31% of trials and spectral coloration complaints in 44% of participants. The structural model's horizontal-only constraint completely eliminates pinna-induced coloration artifacts.

E. Performance Metrics

Single-file processing latency	Value 1.2 - 1.4 seconds	Condition to WAV, 44.1 kHz
4-stem scene render latency	3.8 - 4.6 seconds	4 tracks, mixed sample rates
Peak headroom margin	0.98 (approx -0.18 dBFS)	Multi-track 4-stem scene
ITD at full lateral pan	~29 samples (0.655 ms)	theta = pi/2, Fs = 44.1 kHz
ILD at 4 kHz, full lateral	approx -10 to -14 dB	Contralateral ear, biquad filter
CPU usage, 4-source scene	Single-core MATLAB ~65%	Intel Core i5-12th Gen workstation
Coloration complaints (T3)	0 / 12 participants (0%)	Non-specialist listener panel

VI. COMPARATIVE ANALYSIS WITH PLUGSONIC

Table 2 compares Spatial8 Studio and PlugSonic [1] across the principal architectural and acoustic dimensions relevant to web-based binaural audio platforms:

Binaural Rendering Method	PlugSonic (Comunità et al., 2022)	Spatial8 Studio (2023)
Coloration Risk	Present for non-individual listeners	None on horizontal plane
FIR Filter Size	256-512 taps per ear per source	0 taps (IIR biquad: 2 muls/sample)
Supported Directions	Full 3D (azimuth + elevation)	Horizontal plane (azimuth only)
Multi-Track Mixing	Single source per session	N-track scene with preset arrangements
Sample Rate Handling	Single fixed rate	Auto-resamples all stems to common rate
Mobile/Navigation	ARKit head tracking on mobile	Static source placement (browser only)
Deployment Dependencies	WebAssembly + 3DTI C++ bindings	No external DSP library dependency
Target Use Case	Cultural heritage soundscapes	Music, gaming, voice spatial mixing

VII. LIMITATIONS AND FUTURE WORK

A. Current Limitations

- Horizontal Plane Only: The 2-factor structural model provides no elevation cues. Sources placed at the same azimuth but different heights are indistinguishable. PlugSonic's full 3D HRTF approach supports elevation at the cost of coloration risk.
- Offline Batch Processing: Spatial8 operates as a file-in, file-out batch processor. PlugSonic supports dynamic navigable audio where source positions update in real-time as a user physically moves through a space. Spatial8 does not yet support dynamic listener movement or head tracking.
- No Room Acoustics: Both Spatial8 and the analytical Brown-Duda model produce dry, anechoic output. Early reflection and reverb simulation are absent. PlugSonic similarly lacks real room impulse response simulation in its web deployment.
- Static Source Positions: Spatial8 does not currently support animated source trajectories within a single rendered scene. All source positions are fixed at render time, unlike real-time interactive systems.

B. Future Directions

- Real-Time Streaming Mode: Implementing the ITD and ILD processing as browser-native audio worklet nodes (AudioWorkletProcessor) would eliminate server round-trips and enable live audio spatialization comparable to PlugSonic's dynamic mode.
- Individualized HRTF Integration: Adding an optional HRTF upload path (SOFA format) would allow elevation cue synthesis for listeners who provide their personalized measurements, bridging the gap with PlugSonic's full 3D capability.
- Head Tracking Integration: Extending the mobile interface with device orientation sensors (similar to PlugSonic's ARKit integration) would enable dynamic binaural rendering as the listener moves, reducing front-back ambiguity.
- Perceptual Formal Evaluation: A formal psychoacoustic study with standardized stimuli (ITU-R BS.1534 MUSHRA protocol) would quantify spatialization quality relative to both PlugSonic and full HRTF convolution baselines.

VIII. CONCLUSION

This paper presented Spatial8 Studio, a web-based platform for binaural audio spatialization that converts standard mono and stereo audio into perceptually localized earphone output without HRTF measurement databases. By implementing the structural binaural model of Brown and Duda (1998) Woodworth spherical head ITD delay and continuous first-order head-shadow ILD filtering Spatial8

delivers effective horizontal sound spatialization with zero spectral coloration artifacts and O(1) per-sample computational complexity.

Compared to the most closely related platform, PlugSonic [1], Spatial8 offers a coloration-free spatialization experience for the horizontal plane, eliminates external toolkit dependencies, and introduces multi-track scene mixing with automatic sample rate harmonization capabilities absent from PlugSonic's design. The trade-off is the absence of elevation rendering and dynamic head tracking, which represent clear directions for future development.

Spatial8 Studio demonstrates that physics-based structural binaural synthesis originally formulated as a research prototype in 1998 remains highly relevant and practically superior to empirical HRTF approaches for the specific, common use case of horizontal earphone audio spatialization in accessible, web-deployed environments. The platform is openly available and continues active development toward real-time streaming and individualized HRTF integration.

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